School of Electrical Engineering & Telecommunications

ELEC1111 Workshop 1

Q1. A specific brand of a hybrid vehicle features a high voltage Hybrid Vehicle (HV) battery pack and a 12 V auxiliary battery. The high voltage battery powers the electric motor, generator, air conditioning, compressor, and inverter/converter. All other automotive electrical devices such as the rear lights, horn and radio are powered from a separate 12 V battery.

The HV battery pack is a Nickel Metal Hydride (NiMH) battery pack consisting of 34 low voltage (7.2 V) modules connected in series. The capacity of the battery is 6.5 Ah.

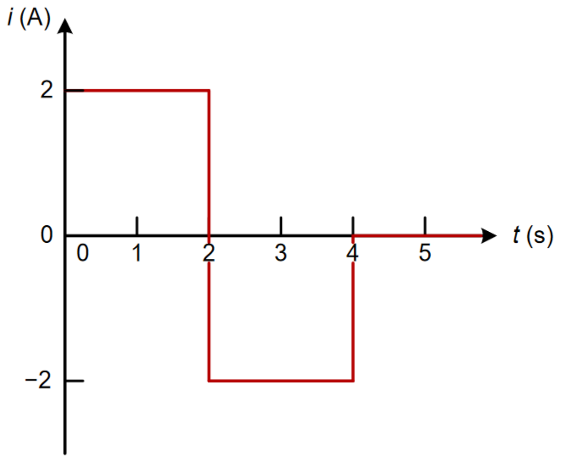
a) How much total energy is available from the HV battery pack? Express your answer in joules and KWh.

Answer: w = 1.591 KWh or w = 5728320 J = 5.728 MJ.

b) If the vehicle drives for 15 minutes as an all-electric vehicle using only the energy available in the fully charged HV battery pack, what is the average current used when driving? And the average power?

Answer: i = 26 A, p = 6364.8 W

Q2. (Solved Examples-Topic 1 Q3) The voltage $v(t)$ across an element is 10 V and the current $i(t)$ through the element is shown in the following figure. Calculate the power and energy and plot their time functions.

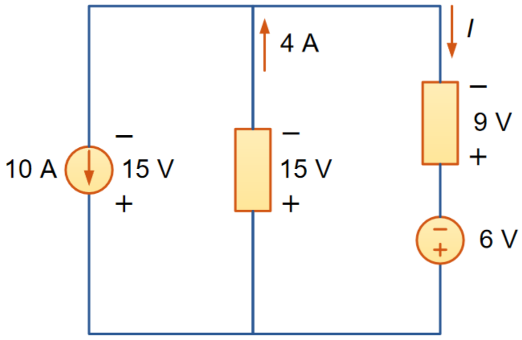


Answer:

$$p = vi = 10i = \begin{cases} 20\text{ W}, & 0 < t \le 2 \\ -20\text{ W}, & 2 < t \le 4 \\ 0\text{ W}, & t > 4 \end{cases}$$

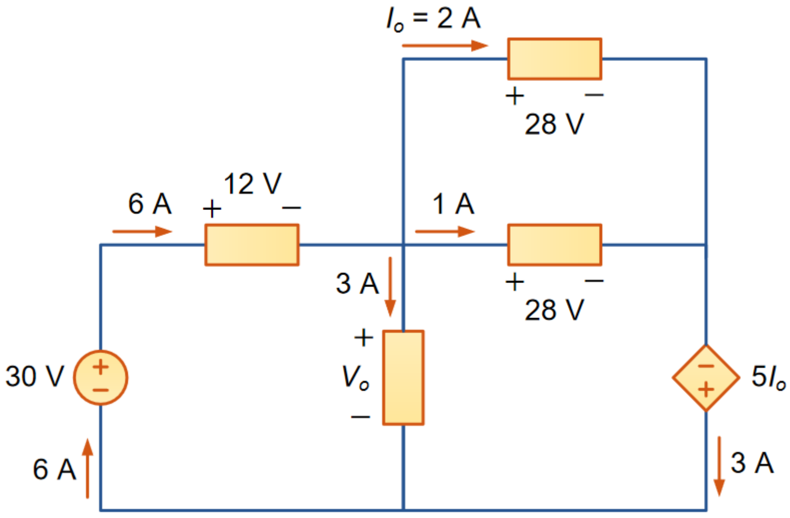
$$w(t) = \begin{cases} 20t\text{ J}, & 0 < t \le 2 \\ -20t + 80\text{ J}, & 2 < t \le 4 \\ 0\text{ J}, & t > 4 \end{cases}$$

Q3. (Solved Examples-Topic 1 Q7) In the circuit below, find the current I and the power absorbed by each element.



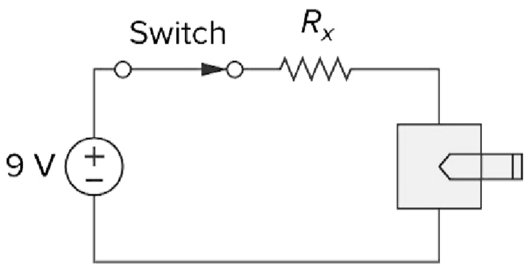
Answer: -6A, − 150 W, 60 W, 54 W, 36 W

Q4. (Solved Examples-Topic 1 Q8) In the circuit below, find Vo and the power for each element.



Answer: 18 V, − 180 W, 72 W, 56 W, 28 W, -30 W, 54 W

Q5. (2019 T3 Midterm Exam Q1) An electric pencil sharpener rated 240 mW, 6 V is connected to a 9 V battery as shown in the figure below. Calculate the value of the series-dropping resistor, Rx, needed to power the sharpener.



Answer: 75 Ω